When Does a Variational Quantum Classifier Head Help?

An Empirical Comparison Against Near-Parameter-Matched Classical Heads on MNIST and CIFAR-10

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Abstract

Hybrid image classifiers that feed a convolutional network into a small variational quantum circuit (VQC) are a common near-term quantum machine learning design, but reported gains are hard to interpret without matched baselines and matched tuning. We train a shared CNN backbone with an 8-dimensional bottleneck and three interchangeable heads (a linear layer, a near-parameter-matched MLP, and an 8-qubit VQC) on low-data subsets of MNIST and CIFAR-10. An exploratory sweep with one shared training recipe suggested a large VQC advantage over the MLP on MNIST (7–18 points) and a large VQC deficit on CIFAR-10 (8–14 points). We re-ran the comparison with longer training, a train/validation/test split and the same validation-based learning-rate search for every head. The mean MNIST advantage shrinks to at most about 3 points (paired over seeds), and the CIFAR-10 deficit to 9.2, 8.9 and 3.1 points at 1,000, 2,000 and 4,000 samples. A linear head has the highest mean accuracy in every condition. We then trained every head on identical frozen embeddings. On embeddings from VQC-trained backbones, a linear readout beats the VQC by 1.6–1.7 points, consistently across seeds. On embeddings from linear-trained backbones, whose values mostly saturate the π tanh angle encoding, the VQC collapses, and so, largely, does a classical Fourier-feature head with the same encoding. Fixed input standardization recovers most of the loss, a learned rescaling recovers a little more, and one tripled training budget recovers little. In this bounded study, the VQC head's behavior tracks classical factors (the training recipe, and the match between the embedding distribution and the encoding), and in no selected configuration does it outperform a linear readout.

1 Introduction

Variational quantum circuits (VQCs) trained by gradient descent are the workhorse of near-term quantum machine learning [1, 2]. In image classification, a common hybrid pattern delegates feature extraction to a classical convolutional network and classification to a small quantum circuit [3, 4]. Reported successes are difficult to interpret: quantum heads are frequently compared against classical baselines with different parameter budgets, different backbones, untuned hyperparameters, or no seed replication. Bowles et al. [8] benchmarked 12 quantum models on 160 datasets and found that simple classical baselines generally match or beat them, and that removing entanglement often does not hurt. Their models read fixed, low-dimensional inputs. We study the image-classification pattern, where a CNN is trained *through* the circuit, and ask how much of a head comparison reflects the head itself, the backbone it induces, and the training recipe.

Our question is narrow: *holding the backbone architecture, the bottleneck dimension and the optimizer fixed, and matching head parameter counts as closely as the architectures allow, does replacing a classical classifier head with a VQC help?* We contribute:

1. An exploratory comparison of linear, near-parameter-matched MLP, and VQC heads (with three data re-uploading schedules) under one shared training recipe on MNIST and CIFAR-10: 90 reported runs, plus two documented unreported CIFAR-10 sweeps (Section 3).

2. A re-run of the head comparison under a train/validation/test protocol with an equal learning-rate search for every head (297 runs, Section 4). Most of the head differences in the exploratory sweep, in both directions, turn out to depend on the shared training recipe: on MNIST largely on its short epoch budget and protocol, on CIFAR-10 largely on its learning rate.

3. A frozen-embedding experiment (498 head fits, Section 5) that separates the head from the backbone it induces. It compares fixed and learned input rescaling with a longer training budget, and includes a classical control that shares the VQC's input encoding.

The negative result is the main finding: in this setting we find no condition in which the VQC head beats a linear readout, and its large apparent advantage over an MLP shrinks to a few points under matched tuning.

2 Architecture and Methods

Shared backbone. All heads use the same backbone architecture and embedding dimension: a CNN backbone ends in global average pooling followed by a linear projection to a $d = 8$ dimensional bottleneck. Each head is trained jointly with its own copy of the backbone, so the embeddings themselves differ (Section 5). For MNIST the backbone has three convolutional blocks (32–64–128 channels, ~94K parameters); for CIFAR-10, four blocks (64–128–256–256 channels, ~0.96M parameters). The backbone is trained jointly with the head in every run; gradients flow through the quantum circuit into the convolutional layers.

Quantum head. The bottleneck vector $\mathbf{x} \in \mathbb{R}^8$ is encoded into an 8-qubit circuit by single-qubit rotations $R_y(\pi \tanh x_i)$ on qubit i. The variational block repeats $L = 4$ layers, each applying learned $R_y(\theta)\, R_z(\phi)$ rotations per qubit followed by a ring of CNOT entanglers. The circuit outputs $\langle Z_i \rangle$ for each qubit, and a final classical linear layer maps the 8 expectation values to 10 class logits. The head stores 154 trainable parameters: $8 \times 2 \times 4 = 64$ rotation angles plus $8 \times 10 + 10 = 90$ readout weights. Only 146 of them can affect the output. The last layer's eight R_z rotations only change the phases of computational-basis amplitudes; what follows them is a CNOT ring, which permutes basis states, and Z measurements, which depend only on amplitude magnitudes. Those phases therefore never reach any $\langle Z_i \rangle$. Parameter counts are used here only to size the baselines; they are not a measure of capacity. Rotation angles are initialized from $\mathcal{U}(-0.1, 0.1)$, which starts every rotation gate near the identity (the CNOT ring is unaffected), instead of PennyLane's default $\mathcal{U}(0, 2\pi)$; in preliminary runs the default produced much larger seed-to-seed variance. In the exploratory sweep, circuits are simulated with PennyLane's `lightning.qubit` statevector backend using adjoint differentiation [9]. Sections 4–5 use a batched PyTorch statevector simulator of the same circuit, which matches PennyLane to 10^{-6} in outputs and in input and weight gradients.

Data re-uploading. We test three encoding schedules. NONE encodes $\mathbf{x}$ once, before the first variational layer. LIGHT additionally re-encodes a rescaled copy ($0.5\mathbf{x}$) midway through the variational block. STRONG places one full encoding of $\mathbf{x}$ immediately before *each* of the $L = 4$ variational layers (the first of these serves as the initial encoding; there is no separate extra encoding), following Pérez-Salinas et al. [5].

Classical baselines. The *linear* head is a single dense layer (90 parameters). The *MLP* head is $8 \to 8 \to 10$: one hidden layer of width 8 with ReLU activation, no dropout or normalization, PyTorch-default (Kaiming-uniform) initialization, and, in the exploratory sweep, the same optimizer and learning rate as every other head (Section 4 tunes each head separately). Its 162 total parameters (weights and biases of both layers) are the closest the architecture comes to the quantum head's 154; we therefore describe the MLP as *near*-parameter-matched rather than equal. The linear head is smaller than both and is included as the strongest simple reference baseline (a linear probe of the embedding), not as a capacity-matched comparison.

Training recipe (exploratory sweep). All runs use AdamW (head learning rate 10^{-3}, backbone 5×10^{-4}, weight decay 10^{-4}), cosine-annealed to zero over the run, batch size 32, and gradient clipping. Training subsets are stratified samples of the full training set, preserving class balance. MNIST runs train for 25 epochs with clip norm 1.0; CIFAR-10 runs for 80 epochs with clip norm 2.0 and standard augmentation (random crop, horizontal flip, rotation, color jitter). Every configuration is repeated with seeds $\{0, 1, 2\}$; we report the mean and *sample* standard deviation over the three seeds. Per-component gradient norms (backbone vs. head) are logged every epoch to monitor trainability.

Table 1: Best test-split accuracy (%, mean ± sample std over 3 seeds) on stratified MNIST subsets. Bold marks the better of the two near-parameter-matched heads (MLP vs. quantum).

			Quantum (re-uploading schedule)		
Train size	Linear	MLP	NONE	LIGHT	STRONG
250	61.5 ± 1.7	36.7 ± 5.4	**54.5 ± 3.7**	48.3 ± 10.7	38.3 ± 7.4
500	79.3 ± 2.9	53.2 ± 6.9	**71.2 ± 1.9**	57.8 ± 2.5	45.6 ± 6.7
1,000	89.6 ± 0.7	76.4 ± 3.4	**83.3 ± 0.3**	68.4 ± 3.5	54.6 ± 14.0

Table 2: Best test-split accuracy (%, mean ± sample std over 3 seeds) on stratified CIFAR-10 subsets. Bold marks the better near-parameter-matched head.

			Quantum (re-uploading schedule)		
Train size	Linear	MLP	NONE	LIGHT	STRONG
1,000	57.5 ± 0.8	**51.2 ± 0.8**	37.1 ± 2.4	34.2 ± 0.7	26.6 ± 1.9
2,000	64.6 ± 0.7	**60.6 ± 0.9**	46.8 ± 5.2	42.7 ± 3.9	28.7 ± 1.9
4,000	71.2 ± 0.5	**67.2 ± 0.4**	58.9 ± 4.6	52.0 ± 7.8	35.2 ± 5.5

Evaluation caveat. In the original sweeps (Tables 1–2), accuracy is measured on the official 10,000-image test split after every epoch and we report the best epoch per run, so the test split participates in checkpoint selection; we call this metric *test-split accuracy*. Mean final-epoch accuracy is within 1.35 points of best-epoch accuracy in all 30 configurations (within 0.5 points in 19), and no ordering among heads changes if final-epoch accuracy is used instead. That comparison does not bound the bias from choices made while looking at these numbers (initialization, augmentation, which sweep to report), so we treat Tables 1–2 as exploratory and re-run the headline comparison under a proper train/validation/test protocol in Section 4.

Sweep provenance. Table 1 is the 25-epoch MNIST sweep; Table 2 is the last of three completed 80-epoch CIFAR-10 sweeps. The two unreported ones differ as follows. One used weight decay 10^{-3} instead of 10^{-4}. The other has identical recorded settings but, judging from its much larger logged train–test gaps, predates the addition of random rotation and color jitter to the augmentation pipeline (augmentation was not recorded in run metadata at the time, so this is an inference). We report the most recent sweep because it reflects the final pipeline. All three give the same ordering of heads at every size (linear > MLP > quantum NONE > LIGHT > STRONG); the unreported weaker-augmentation sweep is *more* favorable to the quantum head (62.9% vs. 58.9% at 4,000 samples). Per-cell numbers for all three are in the released logs.

3 Exploratory Sweep: One Shared Recipe

MNIST: under the shared recipe, the quantum head beats the MLP. Table 1 shows the VQC head (NONE) exceeding the 162-parameter MLP by 17.8, 18.0, and 6.9 points at 250, 500, and 1,000 training samples, while trailing the 90-parameter linear head by 6–8 points. Most of this advantage does not survive the protocol of Section 4. Moving to that protocol (50 instead of 25 epochs, validation-based checkpoint selection, the PyTorch simulator) at the same shared learning rate already reduces it to +10.8, +6.1 and −0.5 points (paired over seeds). Selecting each head's learning rate on validation then gives −2.1, +3.3 and +0.8. We cannot separate the effects of the individual recipe changes.

CIFAR-10: under the shared recipe, the ordering inverts. On CIFAR-10 (Table 2) the VQC head loses to the MLP at every training-set size, by 14.1, 13.8, and 8.3 points, and to the linear head by 20.4 points at 1,000 samples and 12.3 at 4,000. This deficit is also largely recipe-dependent: under the protocol of Section 4, moving the VQC from the shared learning rate to its validated one gains 6.0, 6.2 and 10.7 points, and its gap to the MLP at 4,000 samples falls to 3.1 points (Section 4). Accuracy changes little over the final epochs of these runs, but under a cosine schedule that decays to zero this does not show convergence, and we do not claim it.

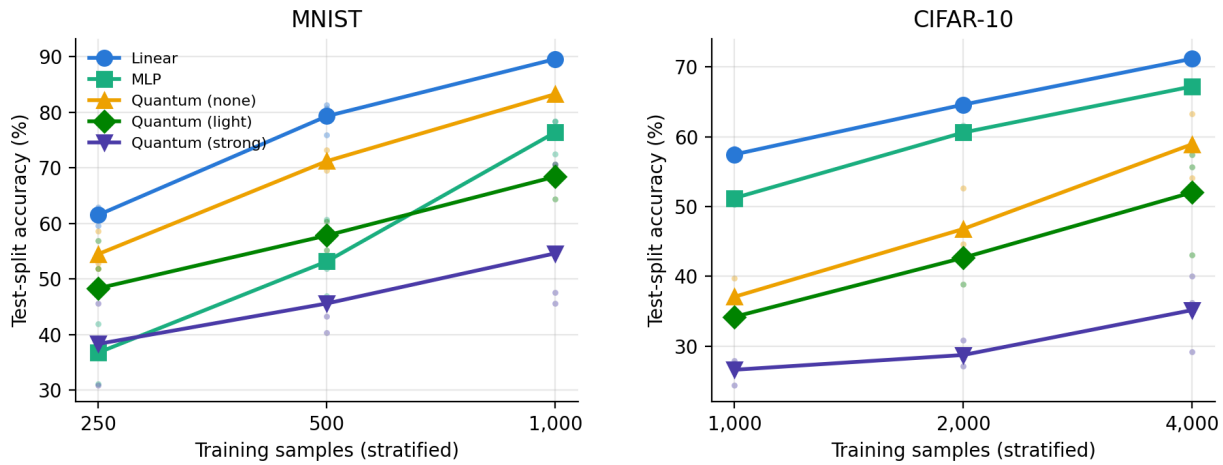


Figure 1: Exploratory sweep (shared recipe; Tables 1–2): best test-split accuracy vs. stratified training-set size. Solid lines show the mean over 3 seeds; faint dots show individual seeds. Left: MNIST. Right: CIFAR-10 (legend shared; note the different vertical scales). The quantum head (NONE) sits between the linear and MLP heads on MNIST but below both on CIFAR-10.

Under the shared recipe, the quantum heads fit their training data poorly. We re-evaluated every saved checkpoint in `eval()` mode on the clean (unaugmented) training subset and the test split; the per-epoch training accuracy in the logs is not comparable, being averaged over a changing model on augmented images in training mode. At 4,000 CIFAR-10 samples the clean train–test gaps are 15.7 (linear), 12.5 (MLP), 8.6 (quantum NONE) and 1.3 points (quantum STRONG), with clean training accuracies of 86.9, 79.7, 67.5 and 36.5%. The strong re-uploading model fits its own training set barely better than the test set; on MNIST it reaches only 58.5% clean training accuracy at 1,000 samples. These are statements about what joint training reached under this recipe, not about representational capacity: at a validated learning rate the NONE head reaches 87.9% clean training accuracy at 4,000 CIFAR-10 samples (Section 4).

The encoding saturates more on CIFAR-10, and saturation tracks seed failures. For each quantum checkpoint we measured the fraction of test-split bottleneck values in the flat region of the encoding, $|\tanh z| > 0.9$. For the 4,000-sample NONE runs it is 16.7, 47.9 and 25.5% on CIFAR-10 (seeds 0–2) and 4.6, 7.5 and 3.2% for the 1,000-sample MNIST runs. Within CIFAR-10, saturation and accuracy are strongly anti-correlated across seeds at fixed training size (Pearson $r = -0.82$ for NONE, -0.88 for LIGHT; 9 runs each): the most saturated seed is the least accurate one at every size. This is correlational evidence, and it does not explain re-uploading, whose MNIST checkpoints show essentially no saturation yet still underperform. Section 5 tests the encoding directly.

Data re-uploading hurts under this recipe. Across all six dataset/size conditions, accuracy orders strictly as NONE > LIGHT > STRONG. On CIFAR-10 at 2,000 samples the strong schedule costs 18.1 points relative to no re-uploading. We do not have a mechanistic explanation for this. Re-uploading enlarges the set of frequencies the circuit can access [6], but that alone does not determine which functions these fixed circuits can realize, and we do not distinguish expressivity from trainability here. It contrasts with the low-dimensional settings where re-uploading is known to help [5]. The re-uploading schedules were not included in the tuned re-run; given how much the NONE head gains from tuning, this result should be read as specific to the shared recipe.

Seed variance. Classical heads are nearly deterministic across seeds on CIFAR-10 (sample std ≤ 0.9 throughout), while quantum heads vary by up to 7.8 points. Given the correlation above, much of this spread coincides with how saturated the jointly trained embedding becomes. We do not interpret it as evidence of barren plateaus [7]: the circuit is shallow, and the gradient norms we log cover the whole head including its classical readout, so they cannot diagnose one. Even the best seed of the best quantum configuration (63.3%) falls short of the mean linear-head accuracy on CIFAR-10 (71.2%). The spread is also recipe-dependent: under the tuned protocol the NONE head's sample std at 4,000 CIFAR-10 samples is 1.1 points (4.6 here).

Table 3: Test accuracy (%, mean ± sample std over 3 seeds) under the train/validation/test protocol with a matched learning-rate search per head. Checkpoint epoch and learning rate are selected on validation only. Selected learning rates: MNIST linear 10^{-1}; MLP-8 3×10^{-2}; MLP-7 10^{-2}, 10^{-2}, 3×10^{-2}; VQC 10^{-2}. CIFAR-10 linear 3×10^{-3}, 3×10^{-2}, 3×10^{-2}; MLP 3×10^{-3}, 3×10^{-3}, 10^{-2}; VQC 3×10^{-3}, 10^{-2}, 10^{-2}.

	Train size	Linear (90)	MLP-8 (162)	MLP-7 (143)	VQC (146 eff.)
MNIST	250	90.3 ± 1.9	78.7 ± 6.5	75.2 ± 1.1	76.6 ± 5.3
	500	94.3 ± 0.2	86.7 ± 3.0	87.0 ± 5.2	90.0 ± 0.6
	1,000	96.4 ± 0.5	93.9 ± 0.8	93.7 ± 0.4	94.6 ± 0.4
CIFAR-10	1,000	59.0 ± 0.9	52.3 ± 1.8	–	43.1 ± 3.8
	2,000	67.6 ± 0.7	62.0 ± 1.0	–	53.1 ± 0.9
	4,000	74.6 ± 0.5	70.8 ± 0.5	–	67.7 ± 1.1

4 Validation Protocol with Matched Tuning

Protocol. For each run we draw a 2,000-image validation set, stratified by class, from the part of the official training set not used for the training subset. The training subset is the same stratified sample as in the exploratory sweep. The epoch is selected on validation accuracy. For each (dataset, size, head) cell, the learning rate with the best mean validation accuracy over seeds $\{0, 1, 2\}$ is selected. We report the test accuracy of the selected checkpoints only; test accuracy for every other grid point is in the released files. Every head gets the same search: the base learning rate (the backbone uses half of it, as before) over $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}\}$ for MNIST and $\{10^{-3}, 3\times10^{-3}, 10^{-2}\}$ for CIFAR-10. For every cell whose selected rate was the largest in the grid, the search was extended by one more step (10^{-1} and 3×10^{-2} respectively). After this extension the selected rate is interior for every MLP and VQC cell. The linear head is still at the largest rate tried on MNIST and at 2,000 and 4,000 CIFAR-10 samples, so its numbers may slightly understate the strongest baseline. MNIST runs train for 50 epochs; CIFAR-10 uses the exploratory recipe (80 epochs, same augmentation). We add the width-7 MLP (143 parameters), whose stored parameter count is closest to the VQC's 146 effective parameters. Re-uploading schedules were not re-run.

Results. Because every head in a cell uses the same seeds, and hence the same training subset and validation split, we compare heads by paired differences (mean ± SD over the three seeds; per-seed values are printed by `summarize_v2.py`). On MNIST, VQC minus MLP-8 is -2.1 ± 9.7, $+3.3 \pm 2.6$ and $+0.8 \pm 1.0$ points at 250, 500 and 1,000 samples, and VQC minus MLP-7 is $+1.3 \pm 4.6$, $+3.0 \pm 4.9$ and $+0.9 \pm 0.7$. The 17–18-point exploratory advantage is gone. The remaining mean difference is at most about 3 points; it is positive for every seed only against MLP-8 at 500 samples and against MLP-7 at 1,000. Three seeds cannot establish that the heads are equivalent, and we do not claim they are. At the shared learning rate of 10^{-3}, VQC minus MLP-8 is already $+10.8 \pm 13.9$, $+6.1 \pm 11.6$ and -0.5 ± 3.8. On MNIST, then, much of the change from Table 1 comes from the combined protocol changes (epochs and schedule length, checkpoint selection, simulator), whose individual contributions we do not isolate. The linear head has the highest mean accuracy in every cell, and among the selected configurations it beats the VQC for every seed. VQC minus linear is -13.7, -4.3 and -1.7 on MNIST and -15.8, -14.5 and -7.0 on CIFAR-10, negative for every seed. On CIFAR-10, where the protocol is held fixed and only the learning rate changes, VQC minus MLP goes from -13.6 ± 1.8, -13.5 ± 2.0 and -10.1 ± 1.6 at the shared learning rate to -9.2 ± 4.5, -8.9 ± 1.7 and -3.1 ± 0.9 at the selected ones. With tuning the VQC also fits its training data far better: 87.9% clean training accuracy at 4,000 samples, versus 67.5% under the shared recipe. Its under-fitting in Section 3 therefore reflects, at least in part, an optimization setting rather than a fixed limitation. Its accuracy varies by up to 12.4 points across the CIFAR-10 learning-rate grid (at 2,000 samples). Its selected rate matches both classical heads' at 1,000 samples, differs from both at 2,000, and matches only the MLP's at 4,000. A head comparison at any single shared learning rate is therefore unreliable.

5 Frozen Embeddings: Separating the Head from the Backbone

In joint training each head induces its own backbone, so a head comparison mixes the two. To separate them we take exploratory-sweep checkpoints of backbones trained jointly with the linear head and with the VQC head (NONE): seeds 0–2, MNIST at 1,000 samples and CIFAR-10 at 4,000. For each we extract the 8-dimensional embedding once, in

Table 4: Test accuracy (%, mean ± sample std over 3 seeds) of heads trained on identical frozen embeddings. Columns give the head the backbone was jointly trained with and, in parentheses, the fraction of test embedding values with $|\tanh z| > 0.9$. Much of the spread within a column is shared across heads (it comes from backbone quality), so differences between heads are reported in the text as paired differences. †Selected learning rate is the largest tried for that head.

	MNIST (1,000 samples)		CIFAR-10 (4,000 samples)	
Head	linear src. (56%)	VQC src. (5%)	linear src. (66%)	VQC src. (30%)
Linear	89.8 ± 0.2	88.9 ± 1.6	70.9 ± 0.6	62.5 ± 4.4†
MLP	88.3 ± 0.9	88.5 ± 1.1	70.2 ± 0.8	62.3 ± 4.0
Fourier	78.2 ± 1.1	87.1 ± 2.0	33.4 ± 1.0	61.8 ± 4.5
VQC	69.0 ± 4.4	87.2 ± 1.0	29.1 ± 3.0	61.0 ± 4.1
VQC+std	81.6 ± 5.6	77.6 ± 6.6	59.5 ± 0.4	53.7 ± 5.6
VQC+std+affine	84.0 ± 3.3	82.7 ± 1.6	65.0 ± 3.1	61.6 ± 4.4
VQC, 3× epochs	–	–	29.6 ± 3.0†	61.0 ± 4.5†

`eval()` mode on clean images, for three sets: the training subset, a 2,000-image validation set drawn as in Section 4, and the test split. We then train fresh heads on these fixed embeddings (AdamW, batch 128, 60 epochs, cosine schedule). Epoch and learning rate are selected on validation. The learning rate is searched in half-decade steps from 3×10^{-3}, and each head's range was extended while its validation accuracy was still rising at the top. That gives maxima of 10 for the two VQC heads without a learned affine, 1 for the linear, Fourier and VQC+affine heads, and 0.3 for the MLP (which diverges at 1) and the 180-epoch VQC. Besides the linear, MLP and VQC heads we train four more:

- a classical *Fourier* head, $[\cos(\pi \tanh z), \sin(\pi \tanh z)] \to \mathrm{Linear}(16, 10)$. It applies the VQC's input nonlinearity without a circuit: $\cos(\pi \tanh z_i)$ is exactly $\langle Z_i \rangle$ after the encoding alone.
- *VQC+std*: the embedding is standardized with fixed training-subset statistics before the unchanged VQC.
- *VQC+std+affine*: the same standardization followed by a learnable per-dimension scale and shift (16 parameters, initialized to the identity).
- the VQC trained for 180 instead of 60 epochs (CIFAR-10 only).

The source checkpoints were selected on the test split in the exploratory sweep, so this experiment is itself exploratory, although all selection within it uses validation only. Because all heads in a column share the same embeddings, we compare them by paired differences over seeds.

On a VQC-trained embedding, the classical heads are slightly better. On embeddings from VQC-trained backbones, the linear head beats the VQC head by 1.7 ± 0.8 points on MNIST and 1.6 ± 0.5 on CIFAR-10 (paired, mean ± SD over seeds). The MLP's margins are 1.3 ± 0.1 and 1.3 ± 0.4. The linear and MLP margins are positive for every seed; they are small, and the marginal standard deviations in Table 4 hide them because most of that spread is backbone quality. The Fourier head is level with the VQC on MNIST (−0.1 ± 1.1, mixed in sign) and slightly ahead on CIFAR-10 (+0.9 ± 0.5). Even on an embedding its own training shaped, the VQC reads it slightly worse than a linear layer.

The VQC-induced backbone accounts for much of the joint-training gap. In the exploratory sweep at 4,000 CIFAR-10 samples the jointly trained linear head beat the jointly trained VQC head by 12.3 points. Two frozen-embedding contrasts bear on this. A refitted linear head reaches 70.9% on the linear-induced embedding but 62.5% on the VQC-induced one, a backbone contrast of 8.4 points. On the VQC-induced embedding the refitted linear head beats the refitted VQC by 1.6 points, a head contrast. Together these account for 9.9 of the 12.3 points. The remaining 2.3 points are associated with refitting heads on frozen embeddings rather than training them jointly, so this is not a complete decomposition, and the split is reference-dependent. With the VQC as the common readout instead, the same four cells give a backbone contrast of −31.9 and a head contrast of +41.8, because the two readouts interact strongly with the embedding source. What the data does show is that a common linear probe scores substantially lower on VQC-trained backbones.

The VQC fails on embeddings whose scale does not match its encoding. Linear-trained backbones produce embeddings with large magnitudes: 66% of CIFAR-10 test values (56% on MNIST) lie in the flat region of $\pi \tanh z$. On these embeddings the VQC collapses. It trails a linear head by 41.8 ± 2.9 points on CIFAR-10 and 20.9 ± 4.4 on MNIST, while linear and MLP heads are unaffected. The classical Fourier head, which shares only the encoding, collapses almost as far on CIFAR-10. It is 4.3 ± 2.0 points above the VQC there and 9.2 ± 3.6 above it on MNIST, so on MNIST the encoding does not account for all of the VQC's loss. Rescaling the input largely repairs the failure. Fixed standardization alone recovers 30.4 ± 2.9 of the 41.8 CIFAR-10 points and 12.6 ± 6.2 of 20.9 on MNIST. Adding the learned affine recovers 35.9 ± 0.9 and 15.0 ± 7.6, so most of the recovery is ordinary preprocessing, and learning the scale adds about 5 points on CIFAR-10. Tripling the training budget recovers 0.5 ± 0.2. The direction reverses on VQC-trained embeddings, which joint training has already fitted to the encoding (5% and 30% saturated). There, fixed standardization *costs* 9.6 ± 7.3 (MNIST) and 7.2 ± 4.9 points (CIFAR-10) relative to the plain VQC. The learned affine undoes this on CIFAR-10 (+0.6 ± 0.3) but only partly on MNIST (−4.5 ± 1.4). These results indicate that the VQC is sensitive to the embedding distribution relative to its fixed encoding, not to normalization as such. They do not isolate scale from centering. Saturation fraction alone does not predict accuracy either: standardizing VQC-trained CIFAR-10 embeddings lowers it from 30% to 12% yet costs 7.2 points. Neither fixed nor learned rescaling is uniformly better than leaving a well-matched embedding alone. Only one longer-training schedule was tested (CIFAR-10 only, at the edge of its learning-rate grid), so these results show that rescaling beats that alternative, not that optimization plays no role.

6 Discussion and Limitations

Interpretation. None of the VQC head's apparent effects in the exploratory sweep survives as an advantage over a well-run classical head. The large MNIST advantage over the MLP came from the shared training recipe as a whole: much of it disappears under the combined protocol changes, and matched learning-rate tuning reduces the mean difference to at most about 3 points. On CIFAR-10, where we can hold the protocol fixed, tuning the learning rate alone shrinks the deficit to the MLP from 10–14 to 3–9 points. In the exploratory checkpoints, a common linear probe scores 8.4 points lower on VQC-trained backbones than on linear-trained ones, more than the 1.6 points by which the VQC reads its own embedding worse than a linear head; this contrast depends on the reference readout, and we did not repeat it for the tuned runs. On a fixed embedding, the VQC reads slightly worse than a linear layer when the embedding suits its encoding, and fails badly when it does not. A classical head with the same $\pi \tanh$ encoding shares much of that failure, and input rescaling largely repairs it. In every selected configuration, a linear readout is at least as accurate as the VQC. This matches the conclusion of broader benchmarks [8] in the specific, widely used setting of a CNN-fed quantum classifier head. For hybrid design it suggests two things. The distribution of the embedding fed to a fixed-range angle encoding should be controlled, by training through the encoding or by rescaling. And quantum heads should be compared against classical heads tuned on the same budget, ideally on the same embedding.

Limitations. This is a bounded empirical study. All quantum results are noiseless statevector simulations. We study one circuit family ($d = 8$ qubits, $L = 4$ layers, one encoding), one bottleneck size and two datasets, with three seeds per cell. That is enough to show consistent signs in paired comparisons, but not to estimate small effects precisely or to establish equivalence. The tuning budget covers only the learning rate. The linear head's selected rate is at the edge of its grid in five of six protocol cells, and some frozen-embedding cells (marked in Table 4) are too. Further tuning there could widen the classical heads' lead, or slightly raise the 180-epoch VQC numbers. Re-uploading schedules were evaluated only under the shared recipe. The frozen-embedding sources are exploratory checkpoints selected on the test split, and they cover only two training sizes. Tables 1–2 use test-split checkpoint selection and should be read as exploratory (Section 2). The mechanism claims concern this circuit and this encoding; whether they extend to other encodings or deeper circuits is untested.

7 Conclusion

Under one shared training recipe, an 8-qubit VQC classifier head appeared to beat a near-parameter-matched MLP on MNIST by up to 18 points and to lose to it on CIFAR-10 by up to 14. With longer training, a validation split and the same learning-rate search for every head, the mean MNIST advantage shrinks to at most about 3 points and the CIFAR-10 deficit to 3–9 points. On identical frozen embeddings a linear readout beats the VQC by 1.6–1.7 points

when the embedding suits the encoding. When it does not, the VQC collapses, a classical head with the same encoding largely shares the failure, and rescaling the input repairs most of it, while more training recovers little. In every selected configuration a linear readout was at least as accurate as the VQC. Claims of benefit from small quantum heads in hybrid classifiers should therefore be made against linear and near-parameter-matched baselines tuned on the same budget, ideally on a shared embedding.

Reproducibility. Code is available at https://github.com/Arkaeyuh/quantcnn. The `results/` directory contains per-epoch metrics and run metadata for all four completed exploratory sweeps discussed in Section 2, the checkpoint and frozen-embedding measurements, and the validation-protocol re-runs; `scripts/make_tables.py`, `analyze_checkpoints.py`, `frozen_heads.py`, `summarize_v2.py` and `make_figures.py` regenerate every number and figure in the paper from those files. The frozen-embedding experiment can be rerun from scratch: the 12 backbone checkpoints it starts from are released in `results/frozen_sources`. Rerunning the clean-image checkpoint analysis (Section 3) needs the other exploratory checkpoints, which are not released; its outputs are. The batched PyTorch statevector simulator used for the new experiments is tested against PennyLane to 10^{-6} in outputs and gradients (`tests/test_torch_vqc.py`). Experiments used Python 3.14, PyTorch 2.11, and PennyLane 0.44 (`lightning.qubit`, adjoint differentiation) on an Apple M3 Max; quantum circuits were simulated on CPU with classical components on the MPS backend. Seeds $\{0, 1, 2\}$ control Python, NumPy, and PyTorch RNGs as well as subset selection.

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